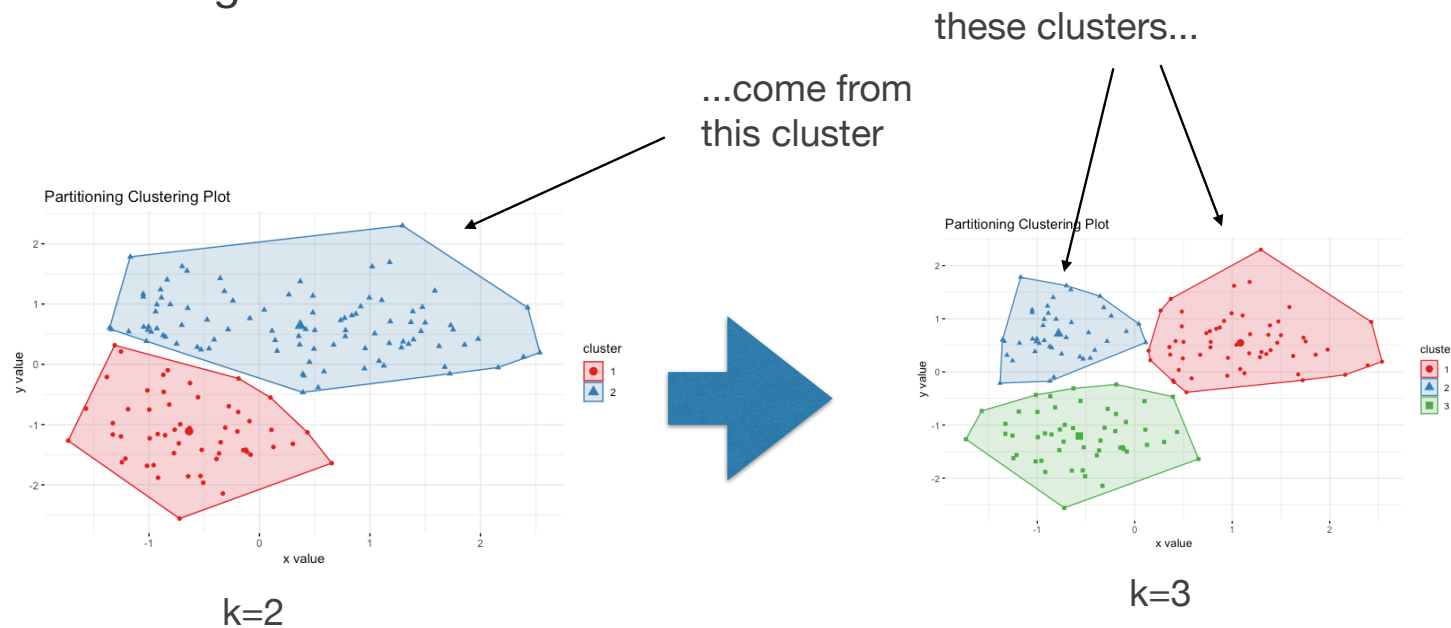


4. Finding structure in the data - hierarchical clustering

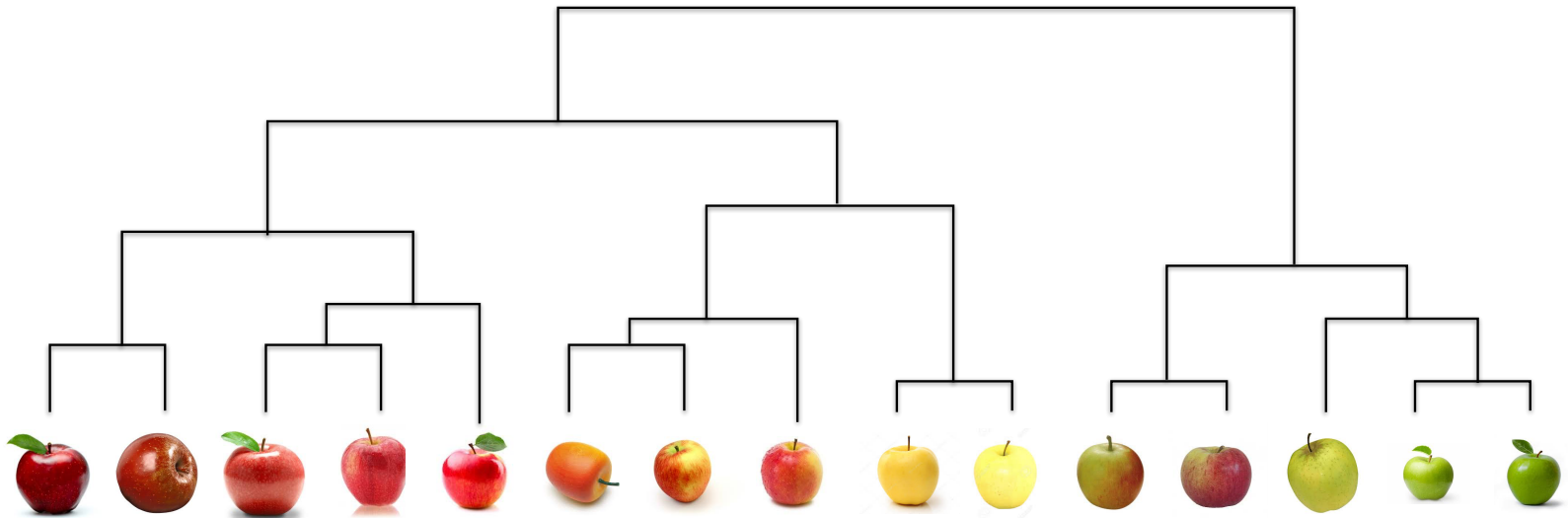
Clustering

- There is often a hierarchical structure in the data that is not captured by the k-means clustering



Hierarchical clustering

- **Divisive clustering** (also "top-down"): iteratively split the full group into sub-groups
- **Agglomerative clustering** (also "Bottom-up"): group iteratively objects from most similar to less similar groups
- tree-like representation of the relationships/distances between objects



Hierarchical clustering

- Starting point: **distance / similarity matrix** between all objects
- Agglomerative** method (“bottom up”)
 - Step 1 : find closest pair (x,y)
 - Step 2 : join objects to form a cluster (x,y)
 - Step 3 : replace x and y in the distance matrix by cluster (x,y) and recompute all distances
 - Step 4 : Repeat from Step1
 - Stop when all objects have been merged!
- Distance between single objects: based on distance measure
- Distance between object and cluster?
 - several possible definitions (= “linkage methods”)

	a	b	c	d	e
a	0	7,1	2,7	5,1	3,8
b		0	3,2	2	1,1
c			0	4,3	4,1
d				0	2,5
e					0

smallest distance

→ join (b,e)



	a	c	d	(b,e)
a	0	2,7	5,1	?
c		0	4,3	?
d			0	?
(b,e)				0

Linkage methods

- Several possible definitions of the distance between two clusters of objects (or an object and a cluster)

- **single-linkage**

$d(A,B)$ = **minimal** distance between all elements of A and B

$$d_{\text{single linkage}} = \min_{i,j} (d(a_i, b_j))$$

- **complete-linkage**

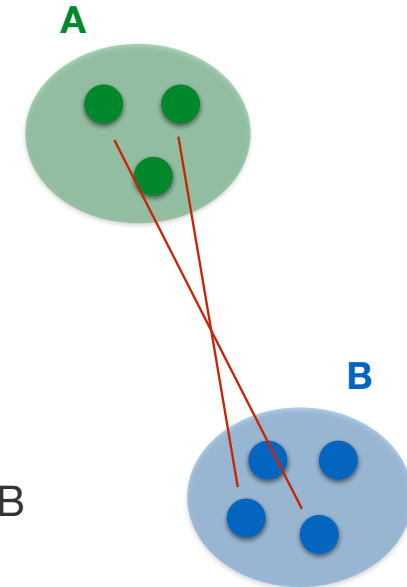
$d(A,B)$ = **maximal** distance between all elements of A and B

$$d_{\text{complete linkage}} = \max_{i,j} (d(a_i, b_j))$$

- **average linkage (UPGMA)**

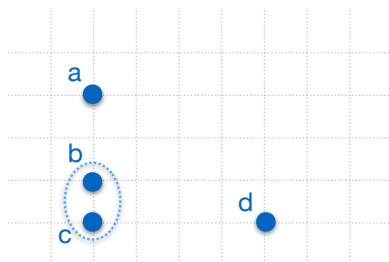
$d(A,B)$ = **average** of all pairwise distances between elements of A and B

$$d_{\text{average linkage}} = \frac{1}{|A||B|} \sum_i \sum_j d(a_i, b_j)$$



Hierarchical clustering

(single-linkage)

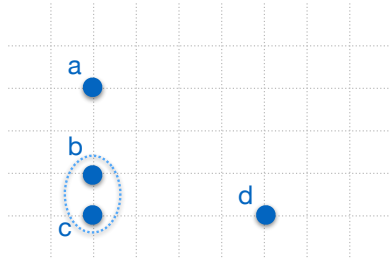


	a	b	c	d
a	0	2	3	5
b		0	1	4.1
c			0	4
d				0

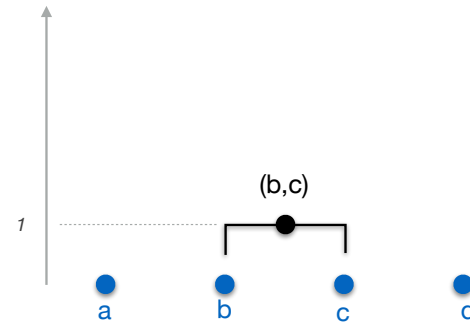


Hierarchical clustering

(single-linkage)

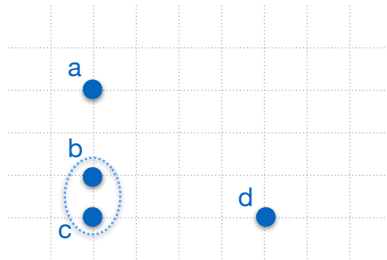


	a	b	c	d
a	0	2	3	5
b		0	1	4.1
c			0	4
d				0

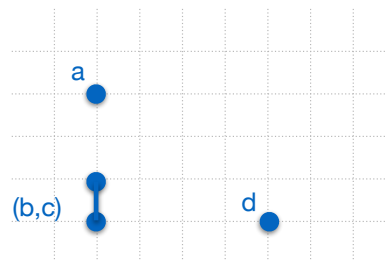


Hierarchical clustering

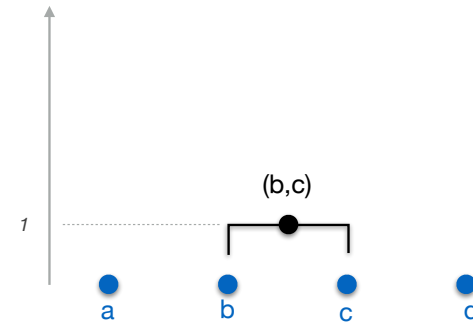
(single-linkage)



	a	b	c	d
a	0	2	3	5
b		0	1	4.1
c			0	4
d				0

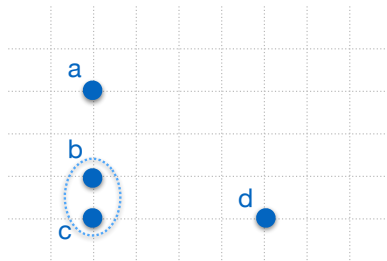


	a	(b,c)	d
a	0	2	5
(b,c)		0	4
d			0

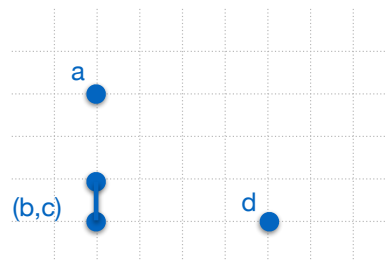


Hierarchical clustering

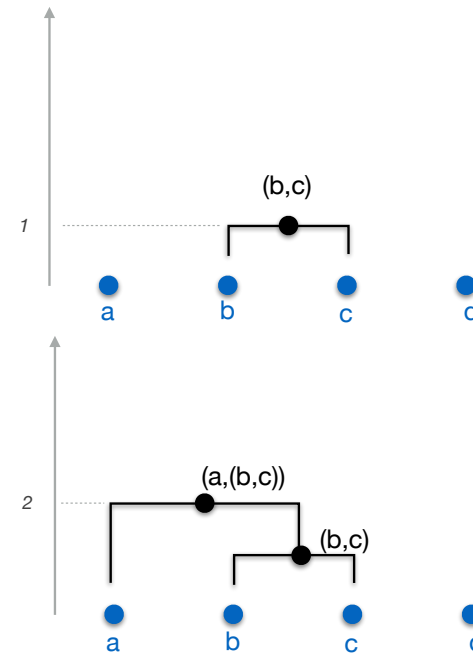
(single-linkage)



	a	b	c	d
a	0	2	3	5
b		0	1	4.1
c			0	4
d				0

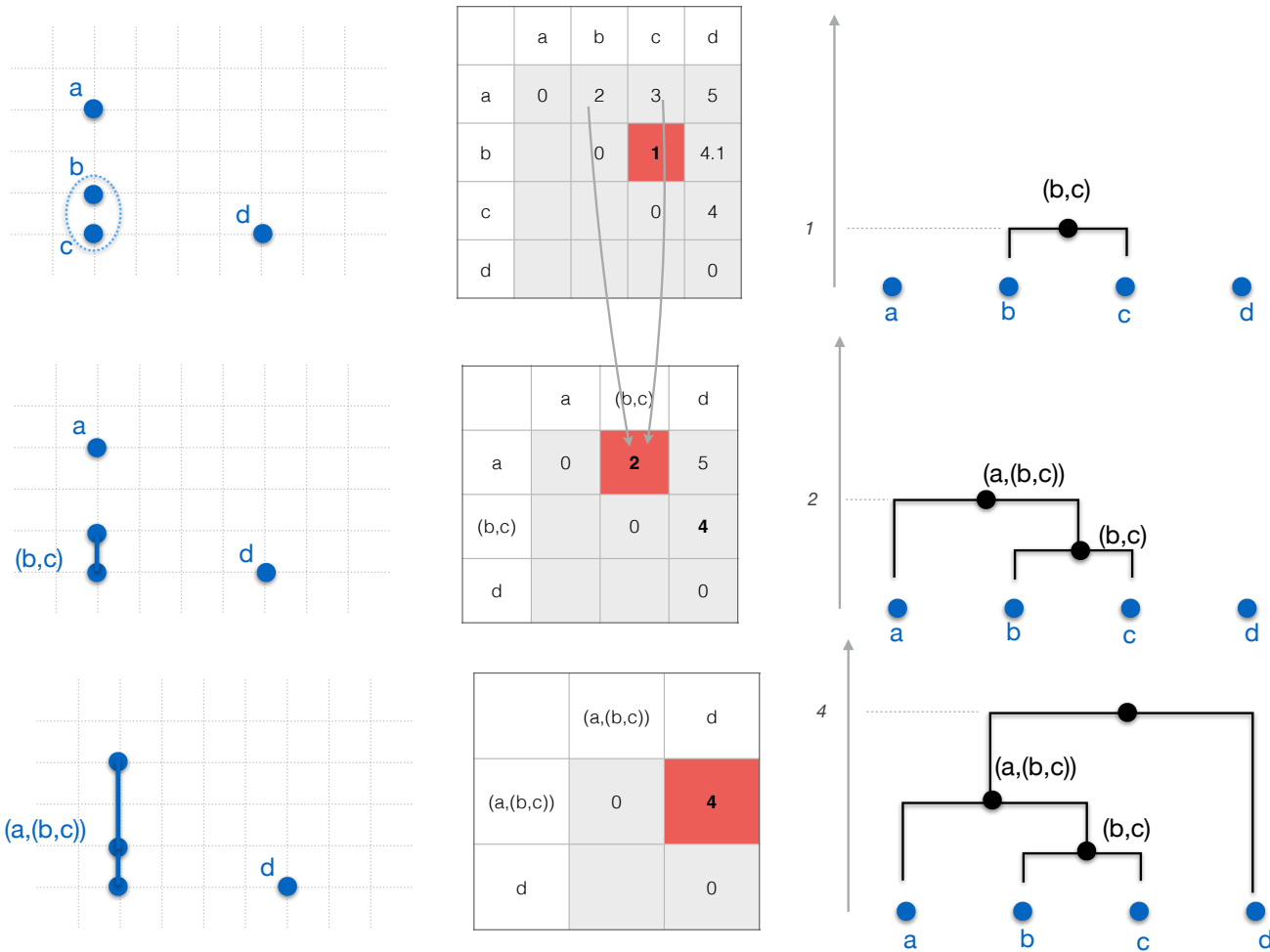


	a	(b,c)	d
a	0	2	5
(b,c)		0	4
d			0



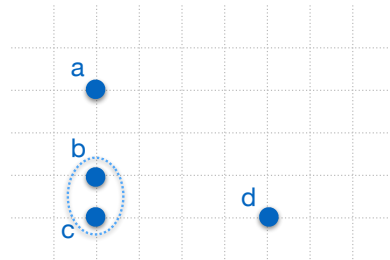
Hierarchical clustering

(single-linkage)

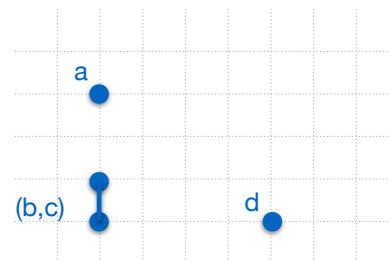


Hierarchical clustering

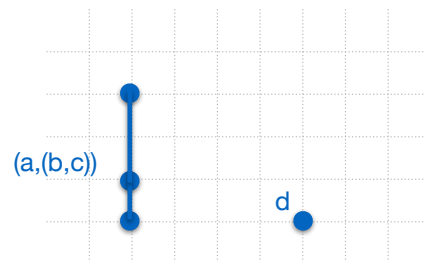
(complete-linkage)



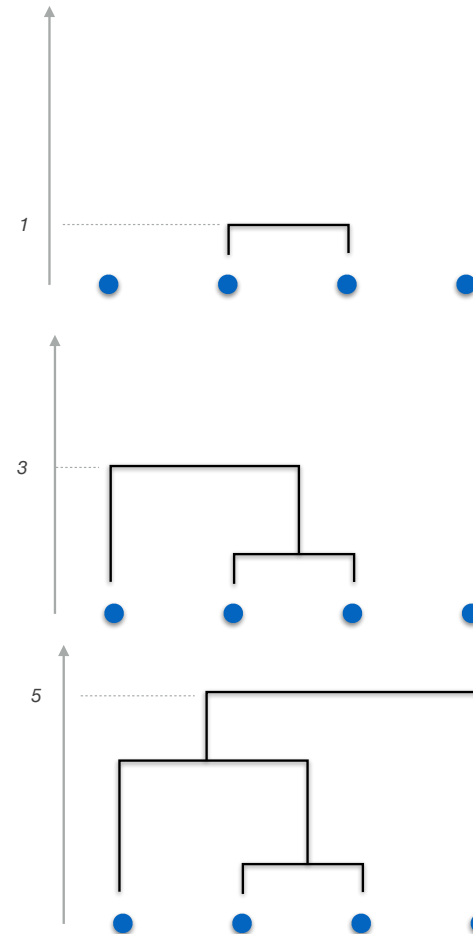
	a	b	c	d
a	0	2	3	5
b		0	1	4.1
c			0	4
d				0



	a	(b,c)	d
a	0	3	5
(b,c)		0	4.1
d			0

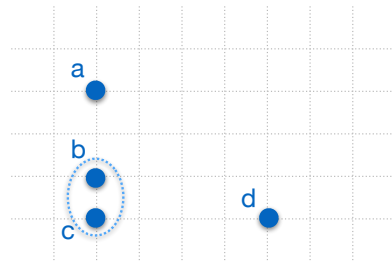


	(a,(b,c))	d
(a,(b,c))	0	5
d		0

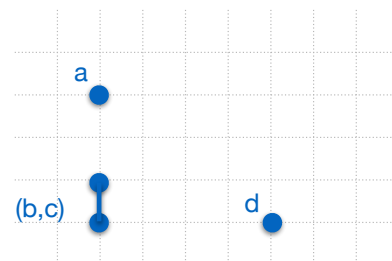


Hierarchical clustering

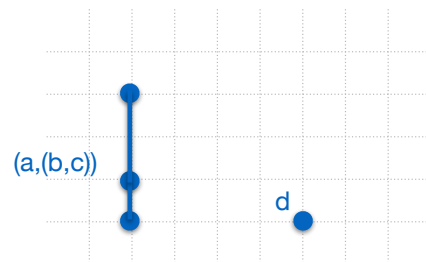
(average-linkage)



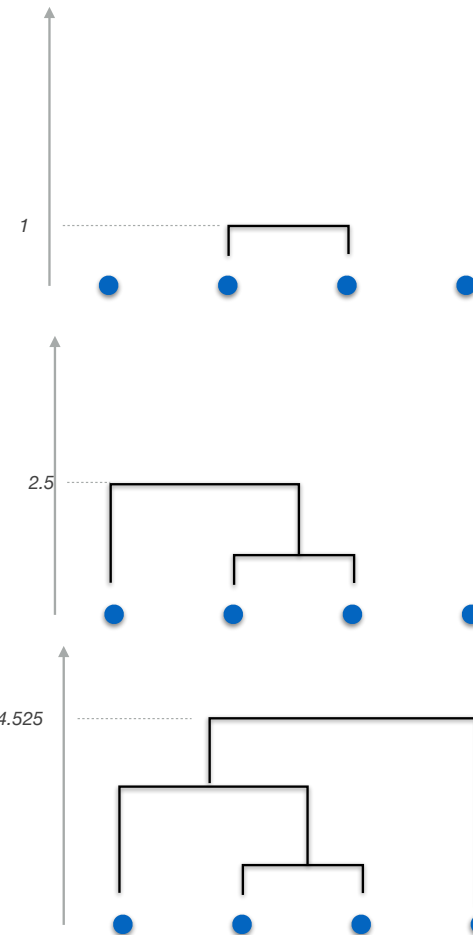
	a	b	c	d
a	0	2	3	5
b		0	1	4,1
c			0	4
d				0



	a	(b,c)	d
a	0	2,5	5
(b,c)		0	4,05
d			0

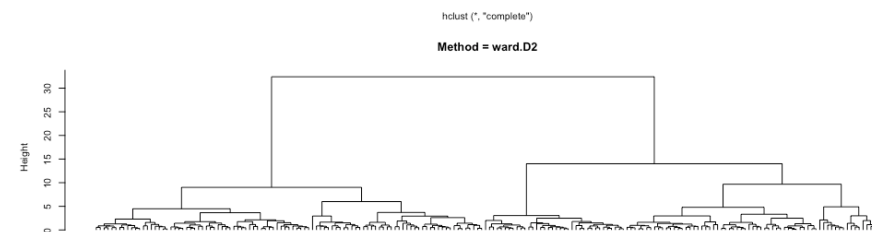
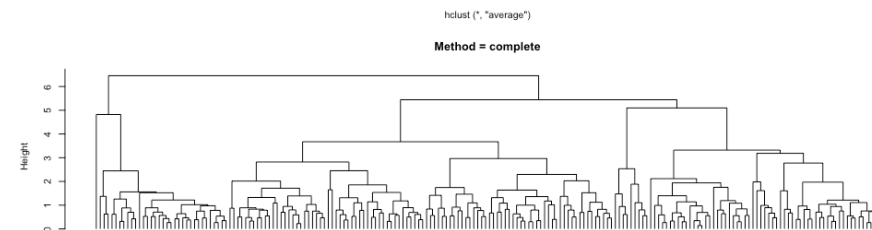
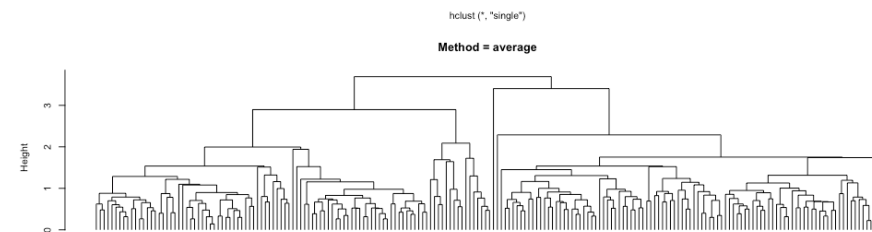
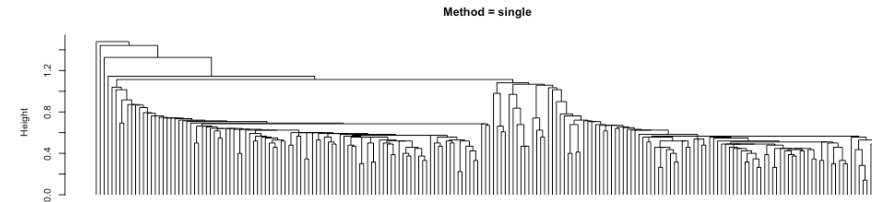


	(a,(b,c))	d
(a,(b,c))	0	4,525
d		0



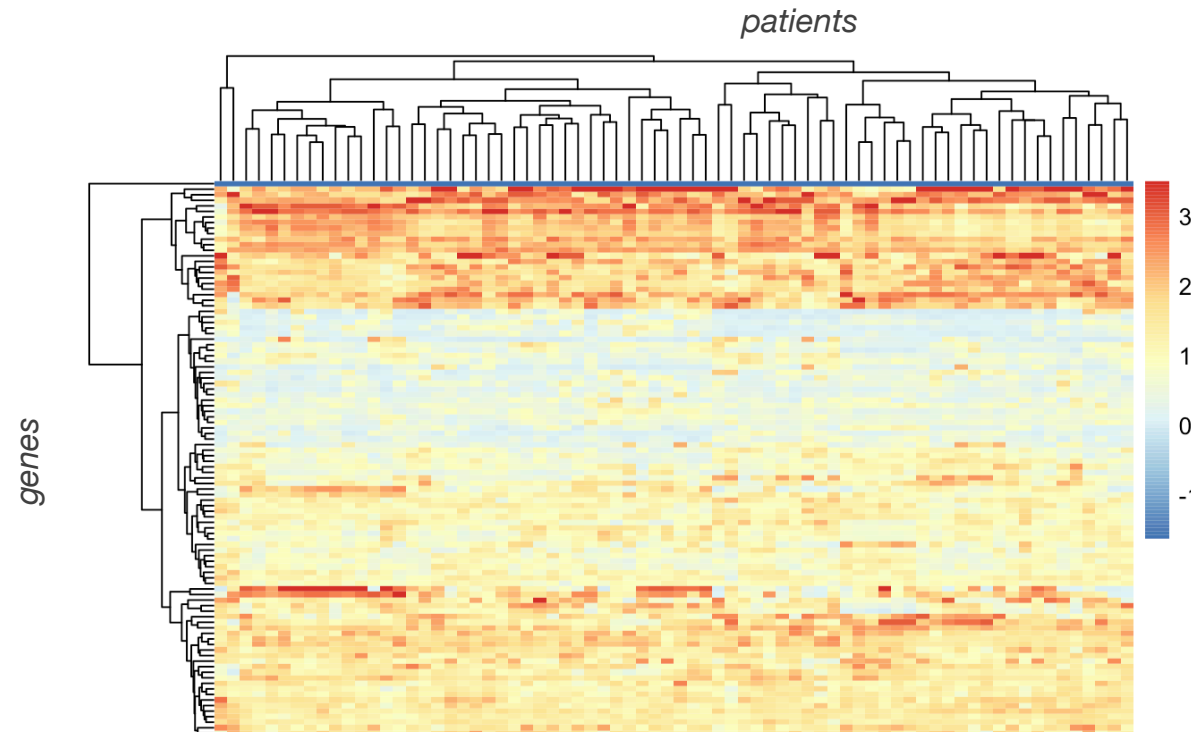
Effect of linkage

- Further linkage methods
 - **centroid-linkage**: distance between the clusters is given by the distance to the centers of the clusters
 - **Ward linkage**: Distance between two clusters is equal to the increase of the variance of the combined cluster compared to individual clusters
 - Each linkage method yields a different tree!

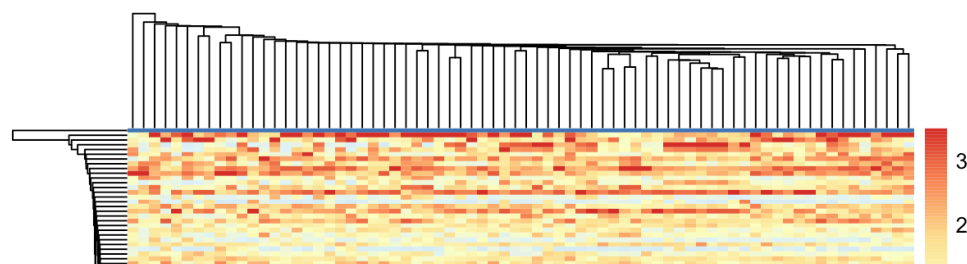


Real world example

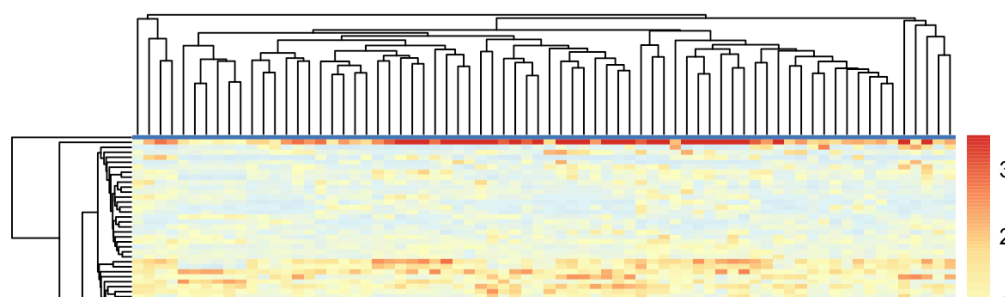
- expression data from 72 AML / ALL patients (Golub et al. 1999)
- clustering of rows and columns: **complete linkage**
- distance between rows / columns: pearson correlation coefficient



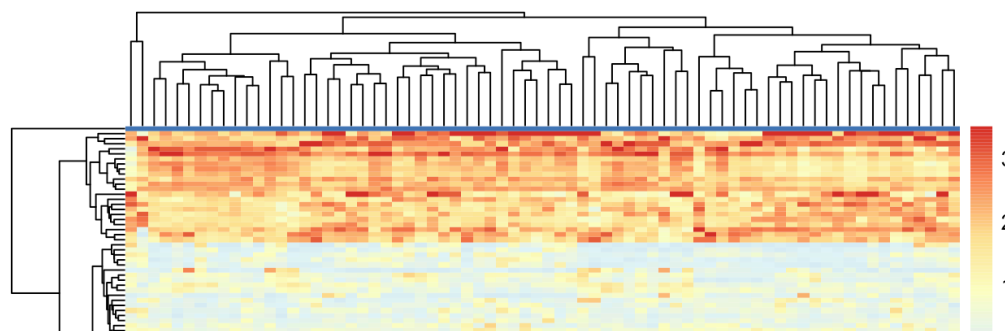
Effect of linkage



single linkage



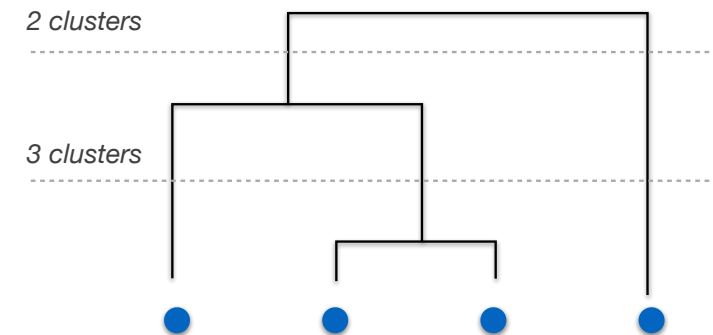
average linkage



complete linkage

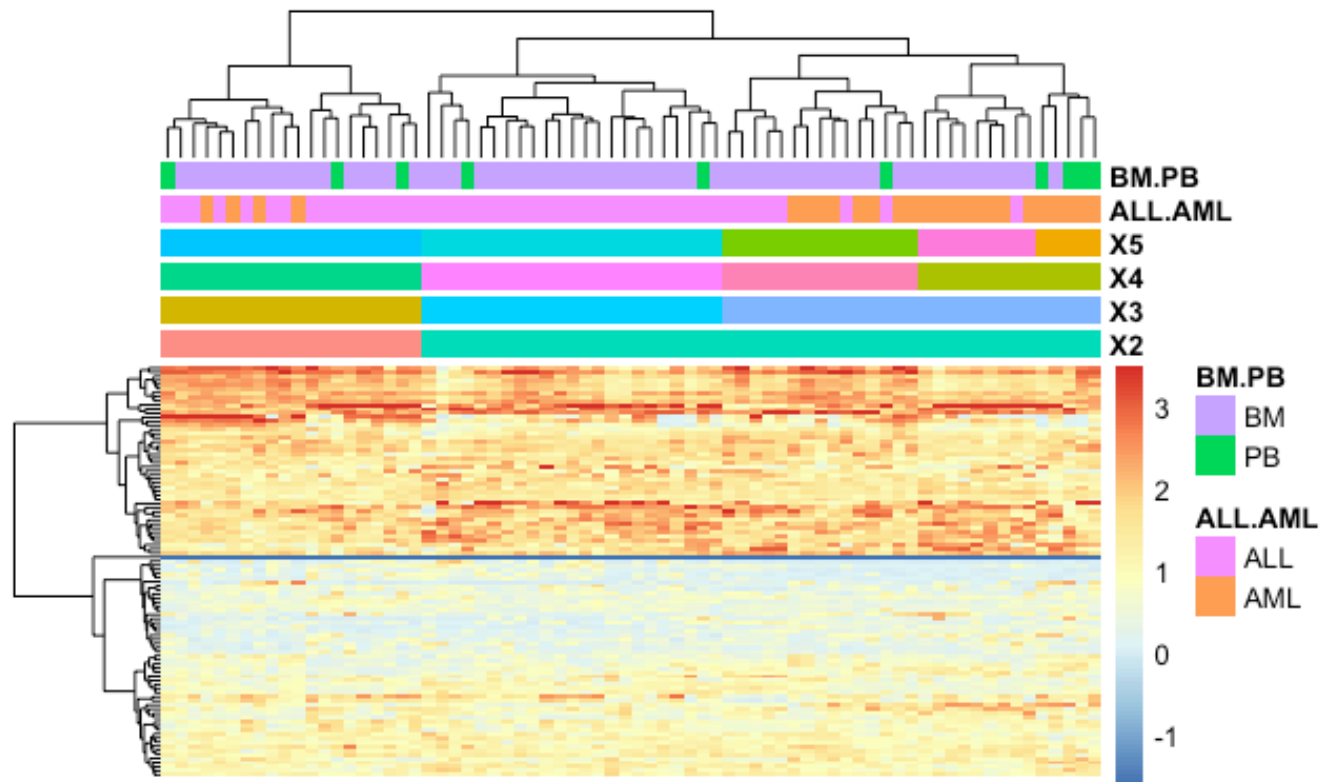
Hierarchical clustering

- Dendrogram (= rooted tree diagram)
- height of a branch = distance between the clusters it joins
- **Different linkage methods** can lead to **different tree topologies!**
- **Different similarity measures** can lead to **different tree topologies!**
- Tree does not yet define clusters:
tree can be cut at different branch height to define clusters
- Where to cut? How many clusters?
→ see criteria discussed previously!
(elbowplot, silhouette,...)



Real world example

Clustering patients

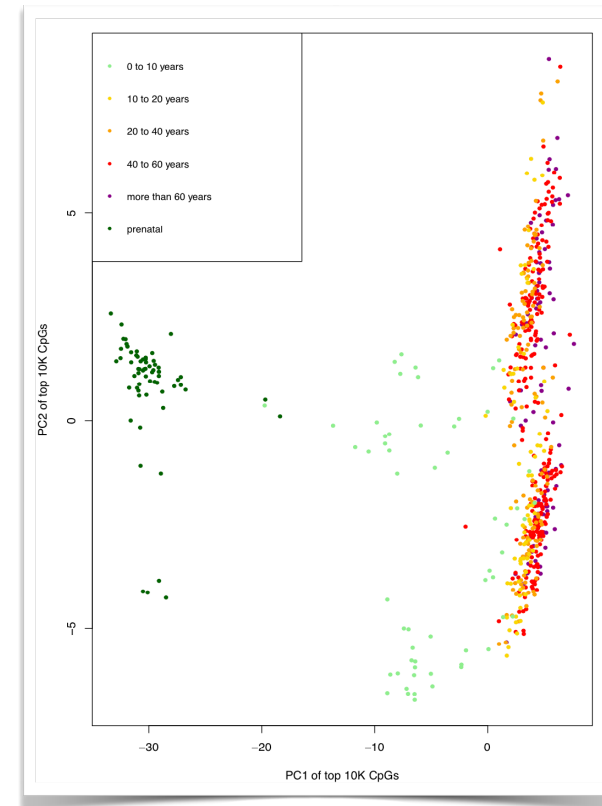


BM.PB: bone marrow or peripheral blood
ALL.AML : acute lymphoid leukemia / acute myeloid leukemia
X2,...X5 : clusters obtained by cutting the tree

4. Finding structure in the data - Principal Component Analysis

Reducing dimensions

- Dataset have a very high dimensionality (e.g. number of genes)
- Need to reduce this large number of dimensions to a smaller number of relevant variables
- Relevant variables = variables which carry most of the information (or variance) of a dataset
- Goal:
 - identify **directions** in the data corresponding to **biological effects**

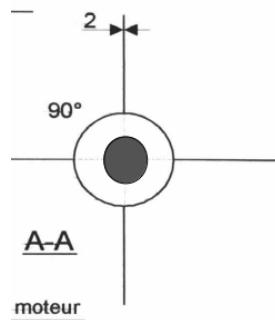


Example of DNA methylation of blood samples
in patient cohort (Jana Dalhoff)
data matrix : 400.000 CpG positions / 250
patients

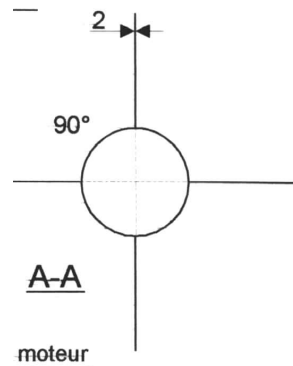
3d → 2d

- How to draw (= 2d) a rocket (= 3d) ?

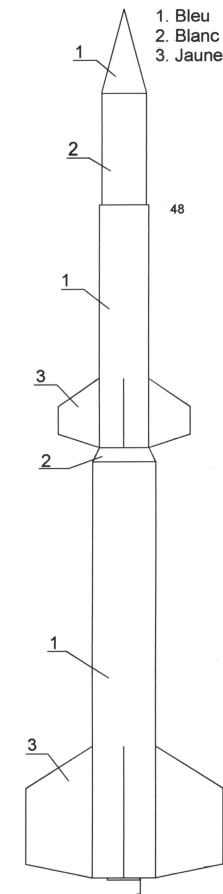
from above



from below



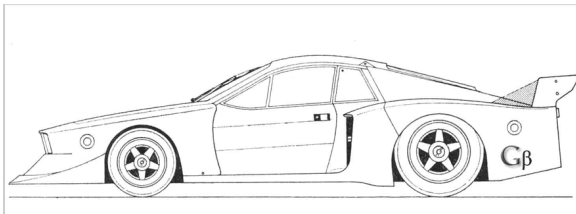
from the side



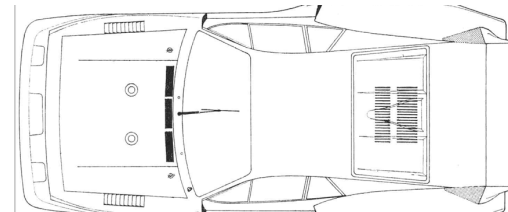
3d → 2d

- How to draw (= 2d) a car (= 3d) ?

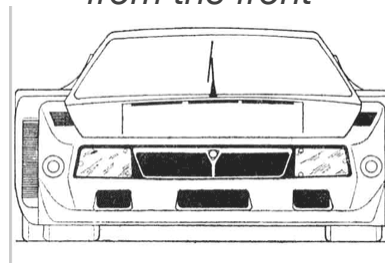
from the side



from above

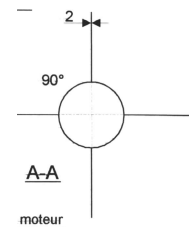
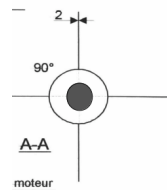


from the front



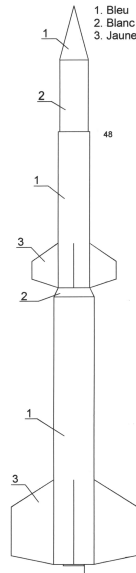
How to choose?

10% information

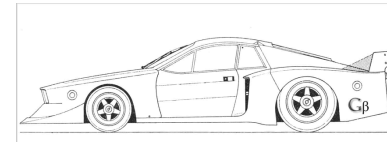


10% information

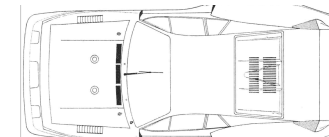
80% information



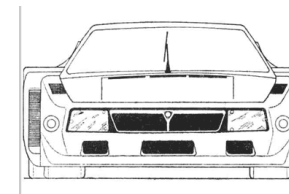
45% information



30% information



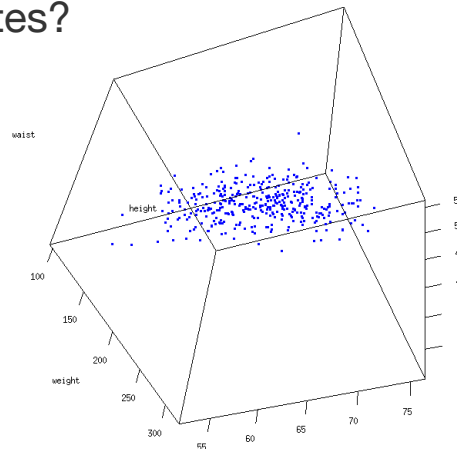
25% information



This is principal component analysis!

Principal Component Analysis

- Idea of principle component analysis: **reduce a multi-dimensional dataset to a lower number of informative dimensions** (“Principal Components”)
- Example diabetes dataset with chol / age / height / weight / waist
- each patient is a dot in a 5d space
→ how to choose new coordinates?

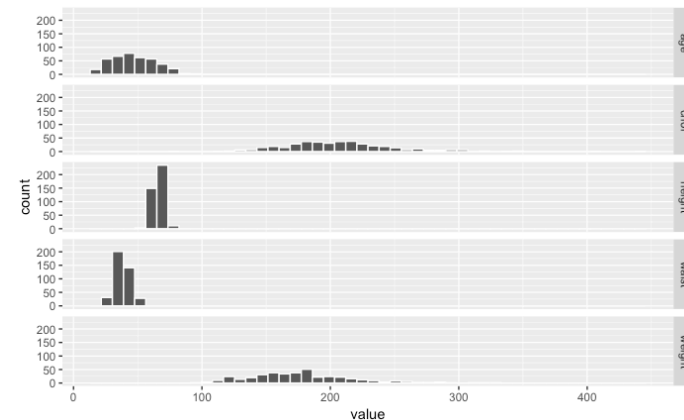


id	chol	age	height	weight	waist
1000	203	46	62	121	29
1001	165	29	64	218	46
1002	228	58	61	256	49
1003	78	67	67	119	33
1005	249	64	68	183	44
1008	248	34	71	190	36
1011	195	30	69	191	46
1015	227	37	59	170	34
1016	177	45	69	166	34
1022	263	55	63	202	45
1024	242	60	65	156	39
1029	215	38	58	195	42
1030	238	27	60	170	35
1031	183	40	59	165	37
1035	191	36	69	183	36
1036	213	33	65	157	37
1037	255	50	65	183	37

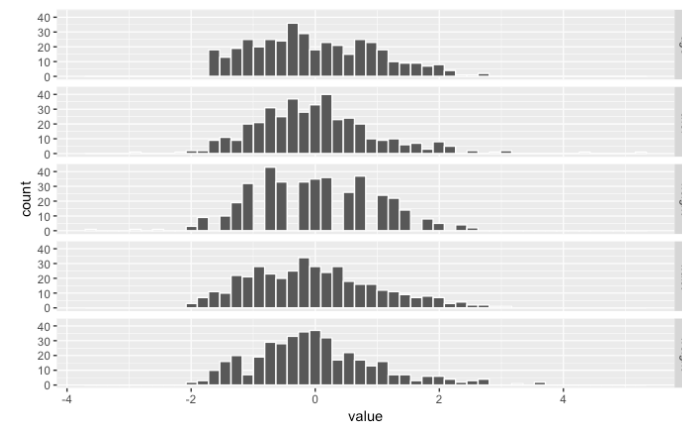
k=17 patients, n=5 variables

Preparing the dataset

Raw data



z-transformed data

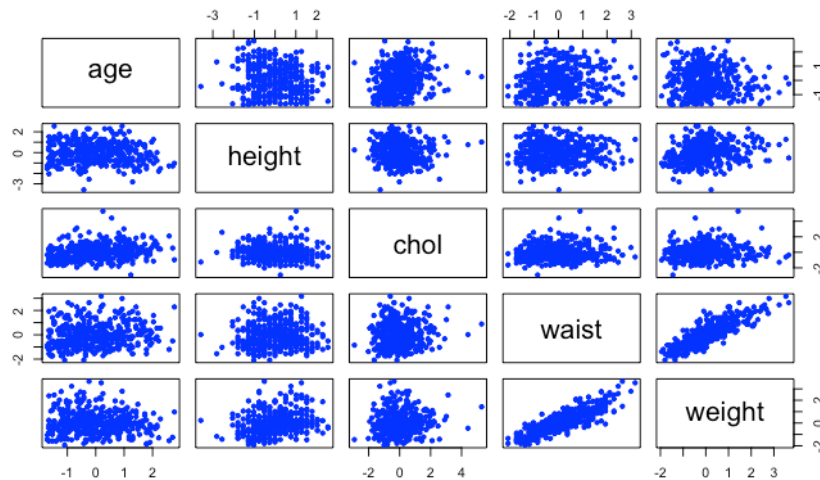


- Each variable has a totally different **range of variation**
- **Solution:** make variables dimensionless using a z-transformation
→ **centering** and **scaling**

$$Z = \frac{x - \bar{x}}{\sigma_x}$$

Correlation structure

- Some variables are highly correlated : knowing one gives a lot of information about the second
- Some are unrelated : no information gain from one over the other



	age	height	chol	waist	weight
age	1,000	-0,095	0,240	0,153	-0,063
height	-0,095	1,000	-0,059	0,057	0,253
chol	0,240	-0,059	1,000	0,112	0,059
waist	0,153	0,057	0,112	1,000	0,850
weight	-0,063	0,253	0,059	0,850	1,000

Correlation structure

- if two variables are **strongly correlated**, they are partly redundant:
knowing the variation of one, you have information about how the second variables changes
→ having 2 variables does not add much information w.r.t. a single variable
- if two variables have **little correlation**, each variable carries information not contained in the other
→ we need to keep these 2 variables to have information about the dataset

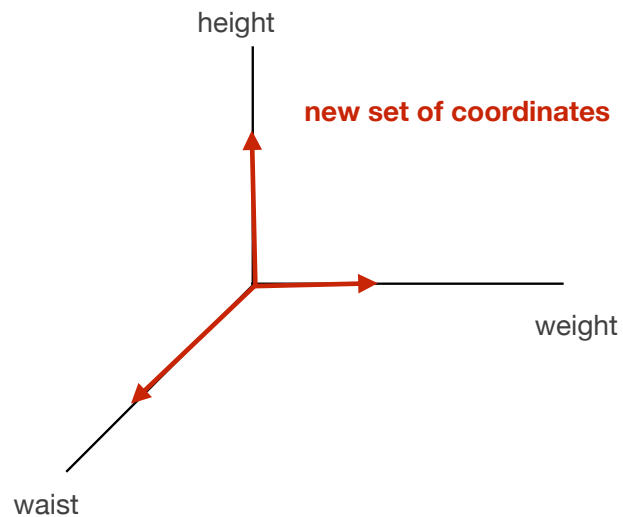


The more diagonal a correlation matrix is, the more information is revealed by the variables

	age	height	chol	waist	weight
age	1,000	-0,095	0,240	0,153	-0,063
height	-0,095	1,000	-0,059	0,057	0,253
chol	0,240	-0,059	1,000	0,112	0,059
waist	0,153	0,057	0,112	1,000	0,850
weight	-0,063	0,253	0,059	0,850	1,000

Defining new variables

- We want to express the dataset in a new set of coordinates
- for each of these coordinate systems, the correlation matrix will change
- **Goal:** find the rotation that makes the correlation matrix diagonal



Hello, matrix diagonalization !

Matrix diagonalization

- For some square matrices X , we can find vectors v such that

$$Xv = \lambda v$$

eigenvectors eigenvalues

- Property: **symmetric matrices** (such as the correlation matrix) are **always diagonalizable**
→ we can find n eigenvalues and n eigenvectors

Matrix diagonalization

1. Consider the correlation matrix A

	age	height	chol	waist	weight
age	1.00000000	-0.09479919	0.23990232	0.15255761	-0.06269027
height	-0.09479919	1.00000000	-0.05853973	0.05661532	0.25298143
chol	0.23990232	-0.05853973	1.00000000	0.11245805	0.05932074
waist	0.15255761	0.05661532	0.11245805	1.00000000	0.84955930
weight	-0.06269027	0.25298143	0.05932074	0.84955930	1.00000000

2. Determine its n eigenvalues and n eigenvectors and build the $n \times n$ matrix V from all the n eigenvectors as columns

```
$values
[1] 1.9201374 1.3081302 0.9011191 0.7635241 0.1070892

$vector
      [,1]      [,2]      [,3]      [,4]      [,5]
[1,] -0.0782536  0.66340112 -0.1957637  0.70124582 -0.15396822
[2,] -0.2139712 -0.41884235 -0.8557896  0.16410053  0.13957979
[3,] -0.1338086  0.59835882 -0.3893703 -0.68726253  0.01108988
[4,] -0.6768556  0.08069999  0.2542954  0.06898591  0.68258976
[5,] -0.6870622 -0.14115337  0.1141285 -0.06508820 -0.70054229
```

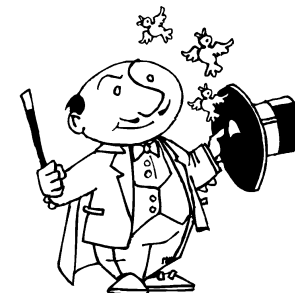
3. Compute

$$S = V' \cdot A \cdot V$$

diagonal matrix!

transposed V matrix

```
      [,1] [,2] [,3] [,4] [,5]
[1,] 1.92 0.000 0.000 0.000 0.000
[2,] 0.00 1.308 0.000 0.000 0.000
[3,] 0.00 0.000 0.901 0.000 0.000
[4,] 0.00 0.000 0.000 0.764 0.000
[5,] 0.00 0.000 0.000 0.000 0.107
```



New coordinate system

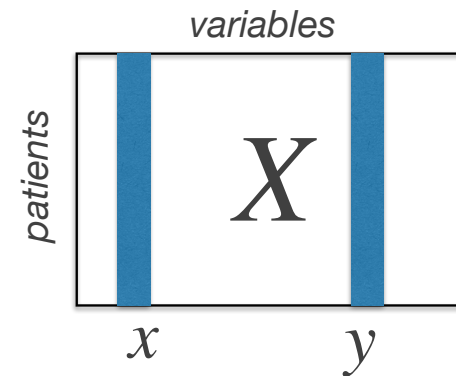
- Correlation / covariance

$$\text{cov}(x, y) = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) \rightarrow \frac{1}{N-1} X'_c \cdot X_c$$

$$\text{cor}(x, y) = \frac{1}{N-1} \sum_{i=1}^N \frac{(x_i - \bar{x})}{\sigma_x} \frac{(y_i - \bar{y})}{\sigma_y} \rightarrow \frac{1}{N-1} X'_{cs} \cdot X_{cs}$$

Z-transformation

c = data centered
 cs = data centered and scaled



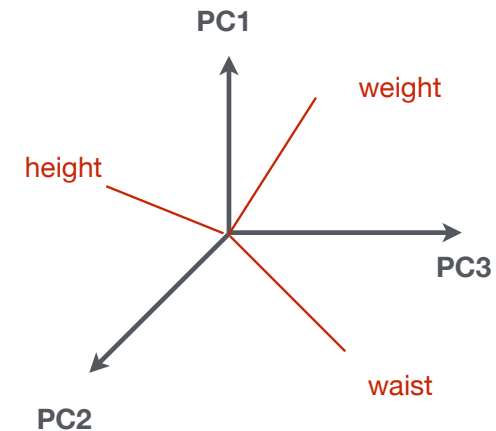
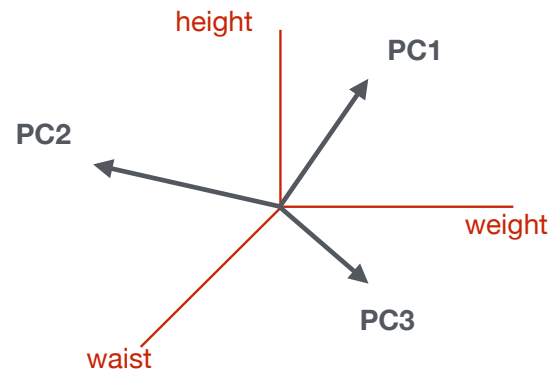
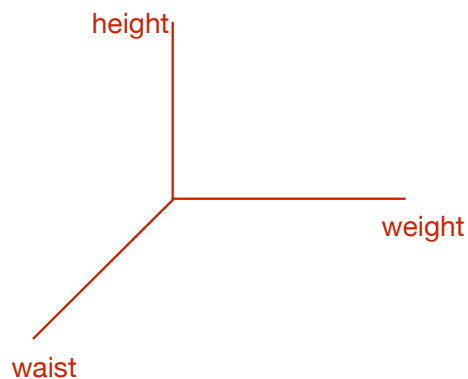
New coordinate system

- V is the **rotation matrix** transforming the initial variables into new variables called principal components

$$S = V' \cdot A \cdot V = V' \cdot \left(\frac{1}{N-1} X'_{cs} \cdot X_{cs} \right) \cdot V$$

$$S = \frac{1}{N-1} (X_{cs} \cdot V)' \cdot (X_{cs} \cdot V)$$

$$V' \cdot X'_{cs} = (X_{cs} V)'$$



New coordinate system

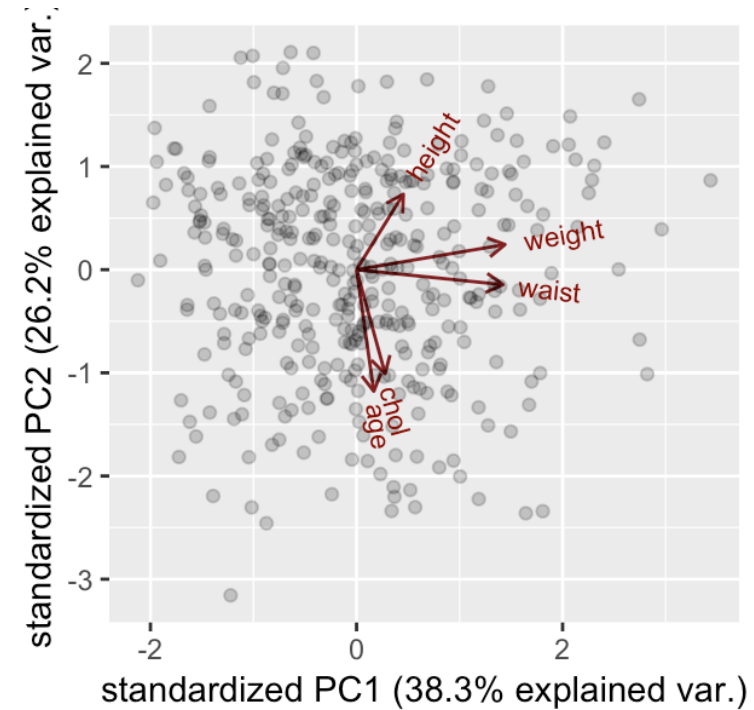
	age	height	chol	waist	weight
[1,]	-0.056039494	-1.023316184	-0.118183417	-1.563795253	-1.418763531
[2,]	-1.090068507	-0.514242224	-0.978388626	1.405952579	0.990046092
[3,]	0.673863338	-1.277853164	0.447741063	1.930025726	1.933703470
[4,]	1.221290463	0.249368716	-2.947805816	-0.865031057	-1.468429708
[5,]	1.038814754	0.503905697	0.923117626	1.056570481	0.120887981
[6,]	-0.785942327	1.267516637	0.900480647	-0.340957910	0.294719603
[7,]	-1.029243271	0.758442677	-0.299279250	1.405952579	0.319552692
[8,]	-0.603466618	-1.786927124	0.425104084	-0.690340008	-0.201942175
[9,]	-0.116864730	0.758442677	-0.706744876	-0.690340008	-0.301274530
[10,]	0.491387630	-0.768779204	1.240035335	1.231261530	0.592716670



	PC1	PC2	PC3	PC4	PC5
[1,]	-2.2724017805	-0.394780391	-0.37314376	0.141536224	0.2090433700
[2,]	1.3055985070	1.119481681	-1.50495414	0.143832964	-0.3513275630
[3,]	2.4741433771	-1.132974842	-1.49880577	0.037584884	0.3143758721
[4,]	-1.8399173774	0.920620520	-0.30773345	-2.959165349	-0.2523118706
[5,]	1.1108365939	-1.098650879	0.71155625	-0.241751636	-0.5571448997
[6,]	0.3019126963	0.582590104	1.33455719	1.004709292	0.1312818611
[7,]	1.2128762055	1.111191009	-0.06294882	0.315416477	-0.9968437164
[8,]	-0.9786990772	-0.575259493	-1.28325003	1.043051917	0.4815398584
[9,]	-0.6156836153	0.831267656	0.56093836	-0.500214895	0.1441444663
[10,]	1.2805031592	-1.405669315	-0.45963453	0.587442537	-0.2560111460

PCA biplot

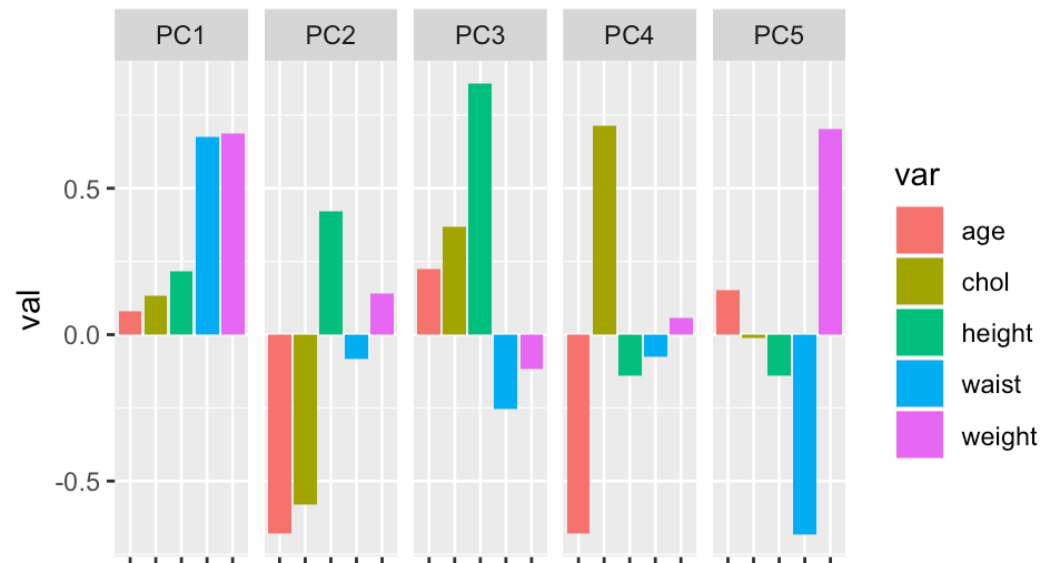
- each **dot** is a sample / patient
- new coordinate system is (PC1,PC2)
- **Red arrows** indicate the contribution of each “old” coordinate to the PCs



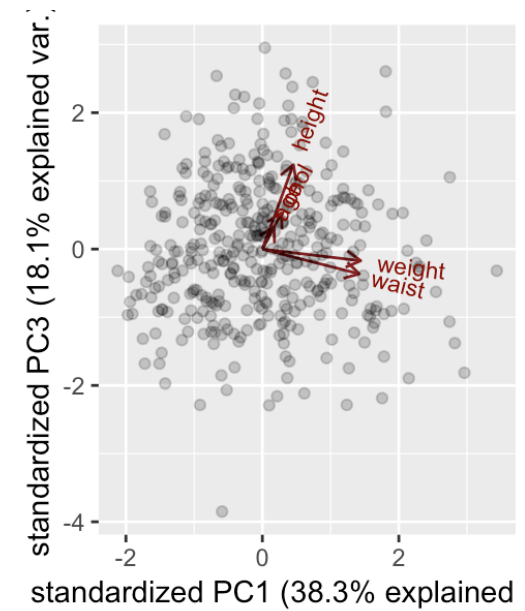
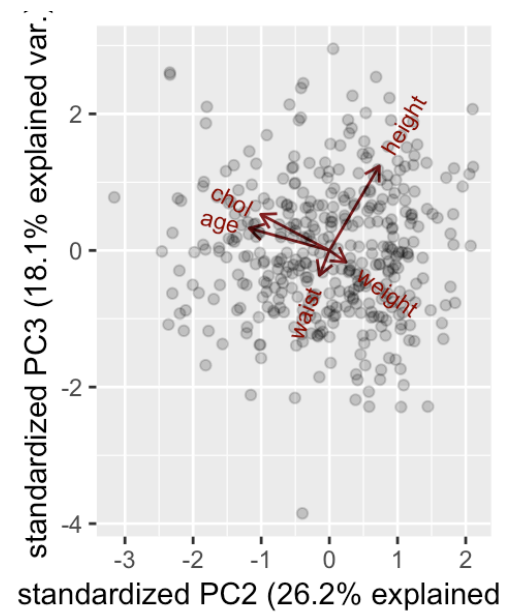
Principal components

- Contribution of each variable to the principal components (coefficients are called "**loadings**")
- Some variables contribute in the same direction to some PCs (e.g. waist and height for PC1), but opposite to others (PC5)

$$PC_i = \alpha_i \cdot \text{age} + \beta_i \cdot \text{chol} + \gamma_i \cdot \text{height} + \delta_i \cdot \text{waist} + \epsilon_i \cdot \text{weight}$$

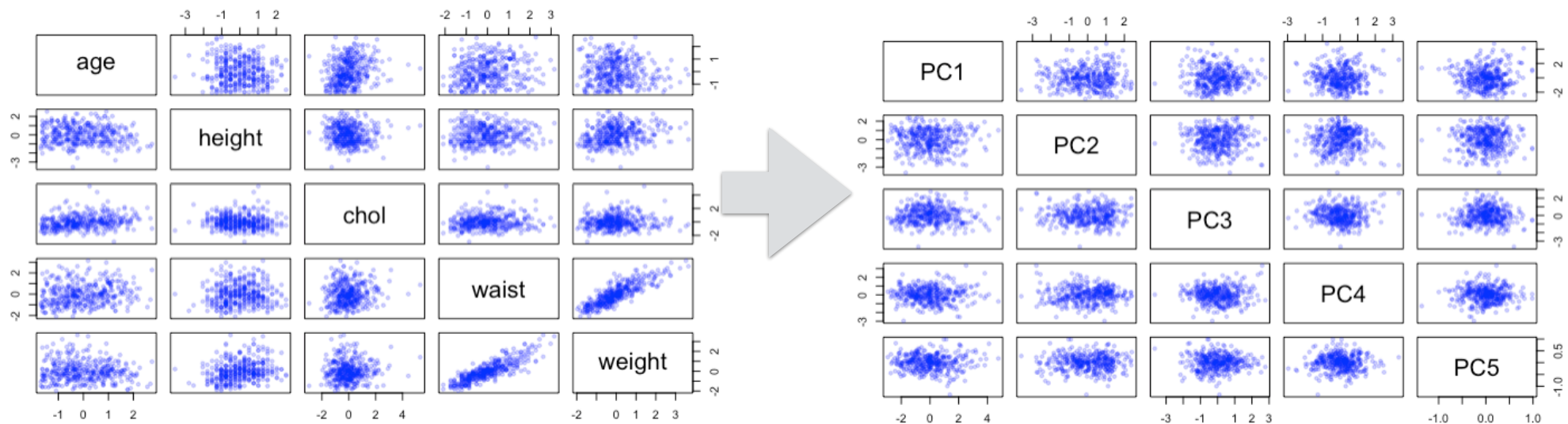


Principal components



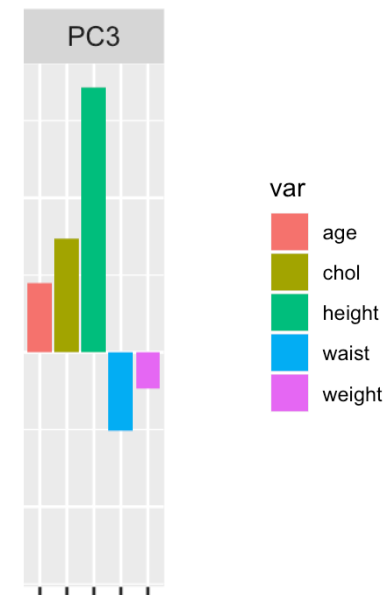
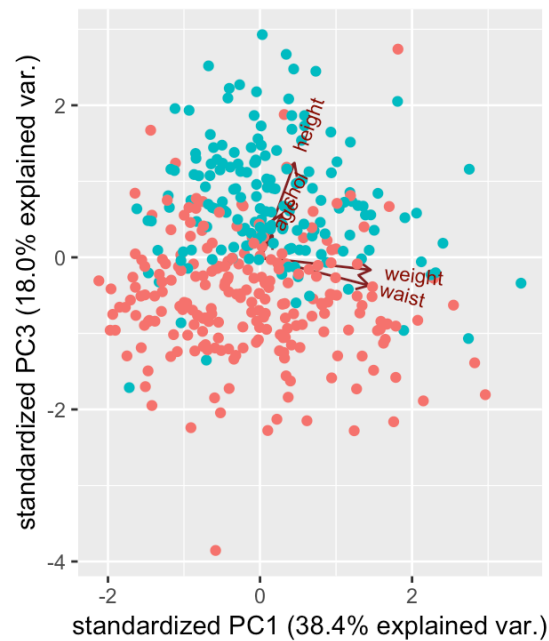
Principal components

- by construction, **Principal Components have no correlation to each other!**



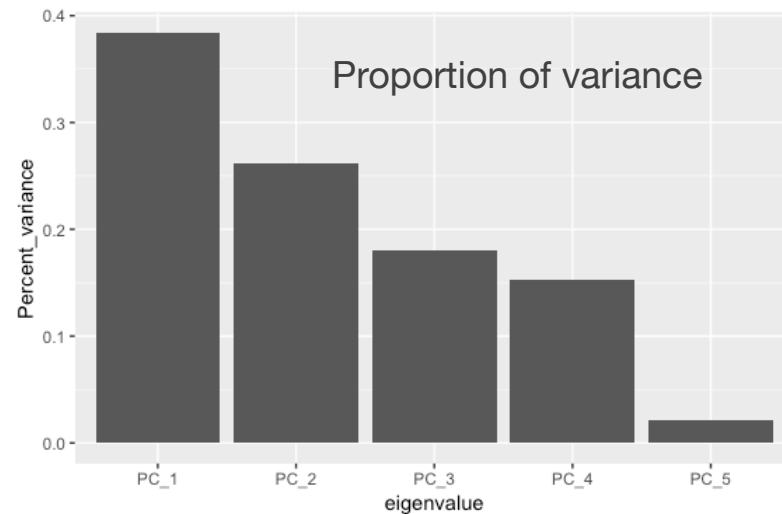
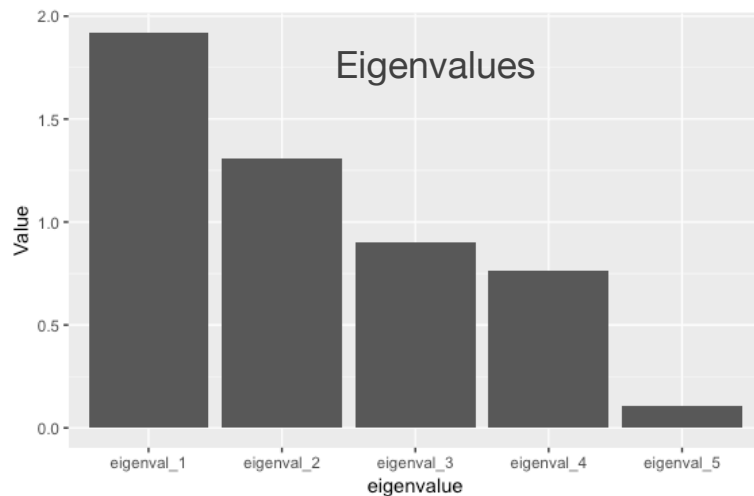
Identifying interesting PCs

- PC plots can highlight a new group structure
- Example: **PC3** seems very associated to gender
- indicates that a combination of height and cholesterol does separate men /women



Number of PCs?

- Each PC explains some part of the **total variance** of the dataset
- This amount is proportional to the corresponding **eigenvalue**
- PCs are ordered by **decreasing eigenvalue** (hence variance)



*Considering PC1 & PC2 explains
63% of the total variance*

Choosing the number of PCs

- several criteria to select the optimal subset of PCs, without losing too much information

- **Proportion of variance:**

keep PCs such that the cumulative variance is above threshold

$$\sum_{i=1}^k \frac{\lambda_i}{\sum \lambda_i} \geq \text{var}_{\min}$$

- **Average eigenvalue criteria:**

keep PCs which have eigenvalue larger than

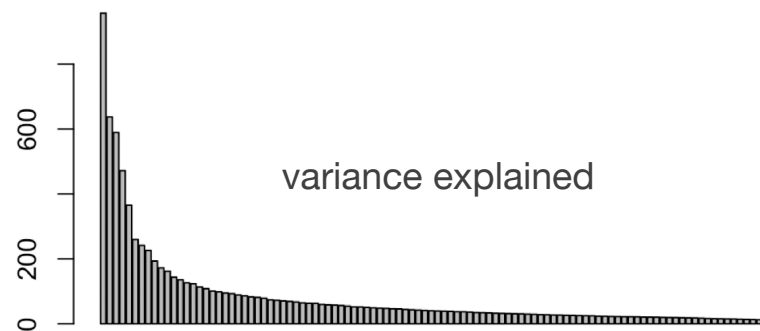
- mean eigenvalue (Kaiser rule) or
- 70% of mean eigenvalue (Jottclife rule)

$$\lambda_i \geq \bar{\lambda}$$

Application to gene expression

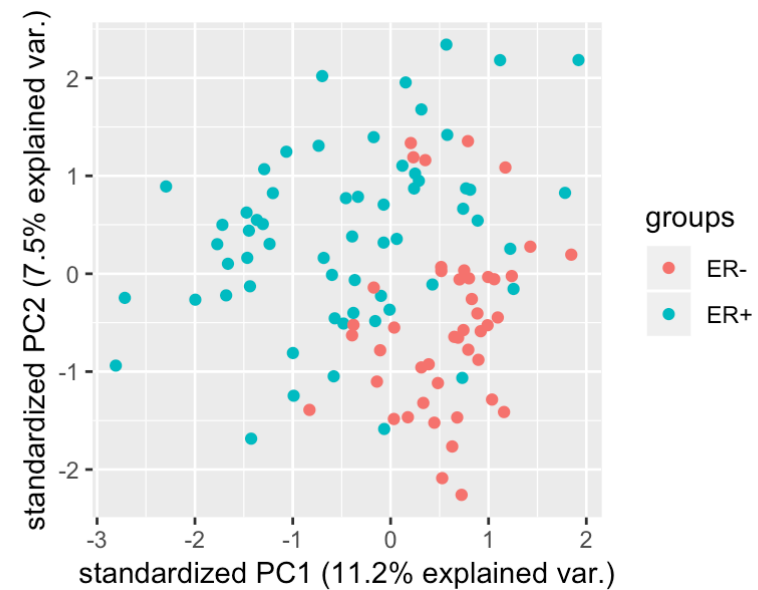
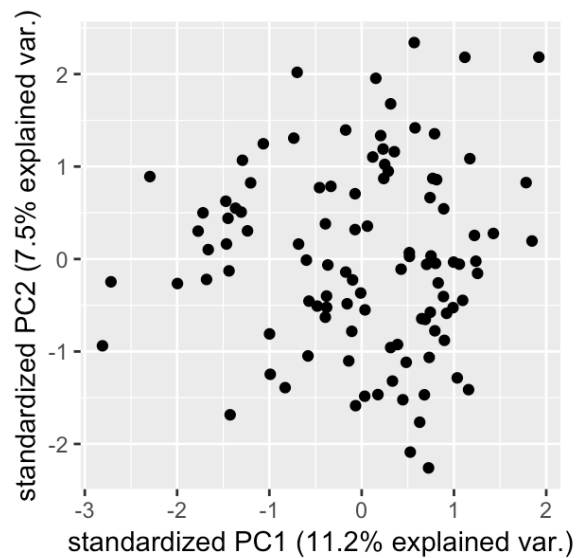
- Gene expression dataset of **breast cancer patients**
- 2 groups: ER+ and ER- patients
- Dimension: $k = 105$ patients / $n = 8534$ genes (here: $n \gg k$)
- pre-processing:
 - **scale** the gene expression across patients
 - **center** the gene expression across patients
- How many principal components do we get?
→ **k**

[M. Ringner, Nature Biotech. (2008)]



Application to gene expression

- PC1 separates ER+ from ER- patients



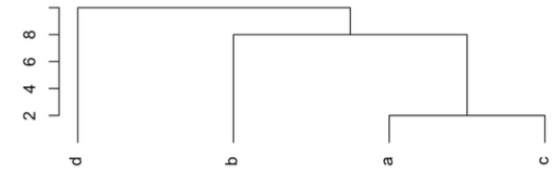
Week 3 - stop and think

*First think about the questions on a sheet of paper;
then use your favorite AI to generate a small R code to illustrate these questions!*

1. Which linkage method was used to obtain the following clustering

- a. single-linkage
- b. complete linkage
- c. average linkage
- d. dendrogram does not correspond to the matrix!
- e. this is not a distance matrix

	a	b	c	d
a	0	4	2	10
b	4	0	8	9
c	2	8	0	3
d	10	9	3	0



- 2. In hierarchical clustering, how would you decide where to cut the tree?
- 3. PCA tries to capture the biggest sources of variation into PC1. Imagine a scenario, in which the real biological information lies in PC2 rather than in PC1...
- 4. On slide 122, how do you interpret that the loadings of some variables are negative in some PCs? What does it mean if 2 variables have opposite loadings in one PC?